

OMIUM × ASCENT HACKATHON · PROBLEM STATEMENT 03

TRUST⁰³CHAIN AI

Autonomous AI-Powered Real-Time **Cybersecurity Protection** Platform

MULTI-AGENT
AUTONOMOUS

REAL-TIME

AUTONOMOUS

// THE PROBLEM

Cybersecurity is reactive, not autonomous.

INCOMING ATTACKS · 24/7

- | | |
|-----------------------|----------------------|
| • Phishing messages | • Scam SMS |
| • Fake payment alerts | • Malicious URLs |
| • Impersonation | • Social engineering |

CURRENT SYSTEMS FAIL BECAUSE

- 01 Require slow manual analysis
- 02 React minutes after damage is done
- 03 No autonomous decision making
- 04 Cannot continuously monitor mobile threats

KEY_INSIGHT *"Users are vulnerable because cybersecurity is reactive instead of autonomous."*

// OUR SOLUTION

Meet Trust Chain AI

An autonomous, AI-powered cybersecurity platform that continuously monitors mobile notifications and neutralizes malicious threats in **real time**.

CORE CAPABILITIES

- 01 Multi-agent AI orchestration
- 02 Android notification monitoring
- 03 Real-time threat protection
- 04 Cloud threat intelligence
- 05 Live SOC dashboard
- 06 Autonomous architecture

DETECTION

<2s

avg threat classify

UPTIME

24/7

always-on monitoring

AGENTS

06

specialized AI agents

REACH

GLOBAL

cloud deployment

// SYSTEM ARCHITECTURE

From device to decision in seconds.



// MULTI-AGENT AI SYSTEM

Six agents. One mission.

AGENT_01

1

Detection

Scans incoming notifications and flags anomalies the moment they arrive.

AGENT_02

2

Threat Intelligence

Cross-references URLs, domains, and senders against live against live threat feeds.

AGENT_03

3

Reasoning

LLM-driven contextual analysis to classify intent, urgency, and tactic.

AGENT_04

4

Response

Issues warnings, blocks payloads, and pushes alerts back alerts back to the device.

AGENT_05

5

Monitoring

Tracks device telemetry and pipeline health across the across the entire fleet.

AGENT_06

6

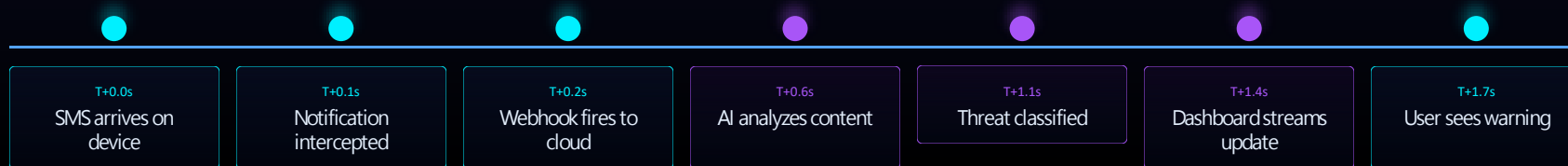
Threat Memory

Stores known attackers, repeat URLs, and adapts the the network over time.

>Agents collaborate via an event bus — every classification feeds the next decision.

// AUTONOMOUS WORKFLOW

Threat in. Defense out.



ZERO_HUMAN_INPUT No analyst. No ticket. No manual intervention required.

// MOBILE MONITORING

Always on. Never asleep.

01 NotificationListenerService

Native Android hook for system-level notification access.

02 Real-time interception

Every push is captured the instant it lands on the device.

03 WhatsApp & SMS coverage

Cross-app monitoring across the highest-risk channels.

04 URL extraction & background sync

Continuous, low-power telemetry that survives app kills.

Trust Chain Monitor

Trust Chain AI Monitor

To enable autonomous threat protection, you must grant Trust Chain AI permission to read notifications so it can intercept SMS and WhatsApp messages before you click them.

Enable Monitoring Service

// LIVE SOC DASHBOARD

A war room in your browser.

- Real-time threat feed

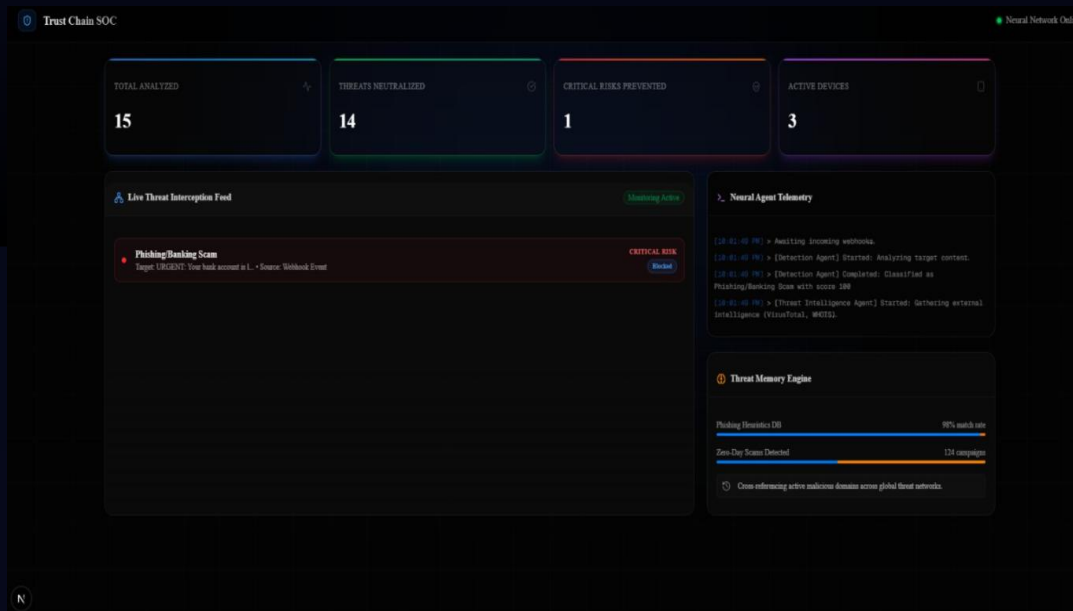
- Telemetry logs

- AI agent activity

- Threat analytics

- Active device monitoring

- WebSocket live updates



// CLOUD DEPLOYMENT

Anywhere there's a signal.

BACKEND

FastAPI

Python async backend deployed across global regions for sub-second latency.

FRONTEND

Vercel Edge

SOC dashboard served from edge nodes for instant instant load worldwide.

REALTIME

WebSocket Mesh

Persistent streams push every event to the dashboard the instant it happens.

EVENTS

Cloud Webhooks

Stateless, signed event ingress from any device on device on the planet.

NETWORK

Cross-Network

Works on Wi-Fi, mobile data, and 5G — no VPN, no VPN, no agent on the carrier.

REACH

Global Coverage

Protection that travels with the user across borders borders and carriers.

DEPLOYMENT_STATUS The platform now works globally.

// THREAT MEMORY ENGINE

The platform remembers.

Every blocked phishing URL, scam template, and malicious domain is etched into long-term memory — so the next attack is recognized before it lands.

MEMORY_INDEX // LIVE

phishing URLs tracked12,480

scam templates clustered3,217

malicious domains flagged8,062

recurring attacker IDs1,944

memory recall latency<80ms

Faster detection

Known signatures resolve in milliseconds, freeing agents for novel threats.

Adaptive intelligence

Reasoning agent uses prior verdicts as context for for ambiguous new payloads.

Evolving defense

Memory grows with the threat landscape — every every device makes the network smarter.

// KEY INNOVATIONS

What makes it different.

01

Autonomous AI security

02

**Real-time mobile
monitoring**

03

**Event-driven
architecture**

04

**Multi-agent
orchestration**

05

Live SOC dashboard

06

AI reasoning engine

07

Threat memory system

08

Global cloud deployment

09

Real-time telemetry

10

Continuous protection

>Each capability is shipping today — not on a roadmap.

// CHALLENGES WE SOLVED

The hard parts, handled.

01 Android notification permissions
System-level access secured with a clean onboarding handshake.

02 Cloud deployment topology
Stateless FastAPI, signed webhooks, zero-trust ingress.

03 Frontend / backend sync
Single source of truth via typed event contracts end-to-end.

04 WebSocket integration
Resilient streams with auto-reconnect and replay buffering.

05 AI reasoning calibration
Tuned prompts and guardrails to minimize false positives.

06 Real-time telemetry pipeline
Sub-second streaming from device to dashboard at scale.

BONUS Autonomous event handling —agents trigger one another with no orchestrator bottleneck.

// FUTURE SCOPE

Where Trust Chain goes next.

ENTERPRISE

SOC integrations

Plug into Splunk, Sentinel, and Chronicle as a first-class signal source.

SURFACE

Browser extension

Real-time URL & form scanning across desktop browsing sessions.

CHANNEL

Email protection

Inbox-side detection that catches spear-phishing before the click.

EDUCATION

AI scam coaching

Inline explanations turn every blocked threat into a learning moment.

GUARDIAN

Elderly protection mode

Family-linked accounts with stricter blocking and trusted alerts.

PLATFORM

Enterprise SaaS

Multi-tenant control plane, RBAC, and per-org threat policies.

NORTH_STAR A global threat intelligence network that learns from every device it protects.

// REAL-WORLD IMPACT

Protection that **reaches everyone.**

Trust Chain AI defends the people most often targeted by digital fraud — and the — and the organizations that can't afford to be breached.

PROTECTS_AGAINST

- Phishing campaigns
- SMS & payment scams
- Social engineering
- Malicious links & payloads

Everyday users

Daily mobile defense

Elderly

Guardian-grade safety

Children

Safer messaging

Enterprises

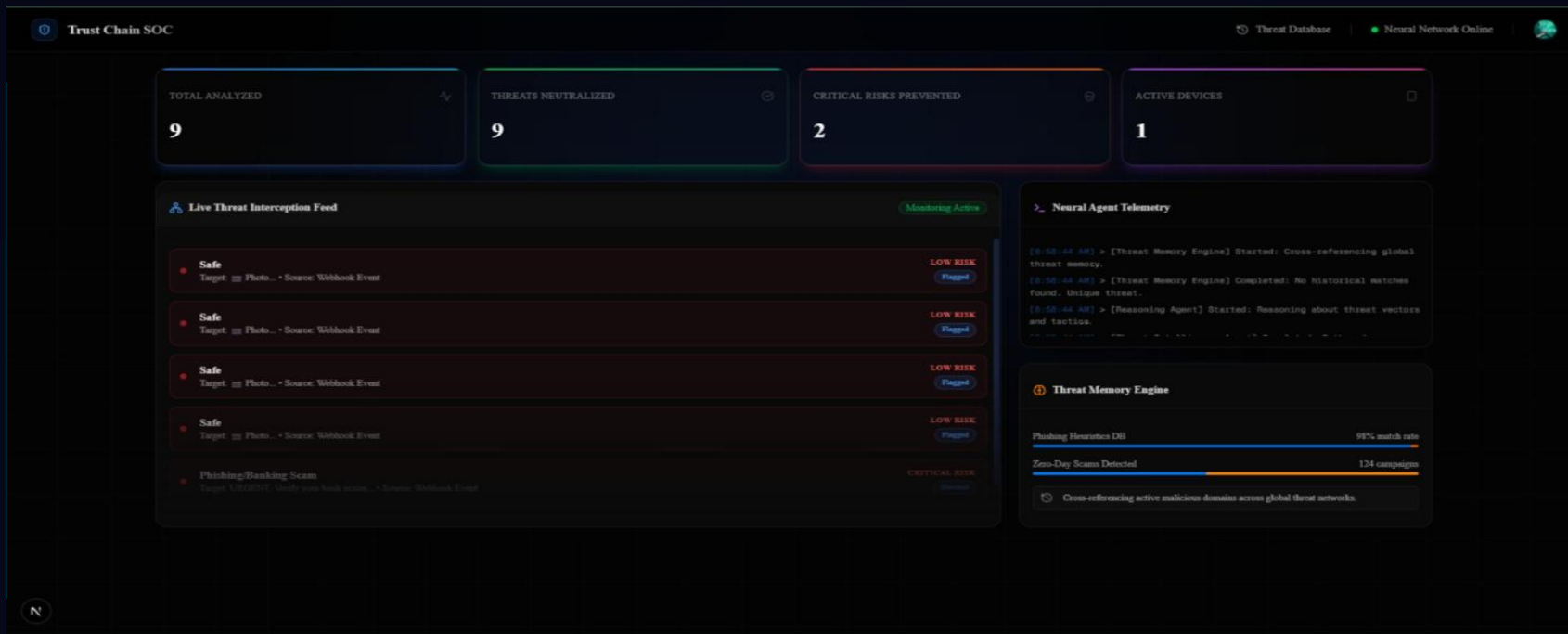
Workforce protection

Organizations

Network-wide shield

// LIVE DEMONSTRATION

See it in action.



THANK you

TRUST CHAIN AI

Autonomous AI Cybersecurity for the Modern World.